Autonomous Elucidation of Homogeneous Organic Reaction Mechanisms through Directed Acyclic Graph Evidence Delivery and Peer-to-Peer Agent Orchestration

Chengyi Yang[1*], Yang Li[1*], Jierui Liu[2], Yuzhou Chen[1], Xirui Cao[1], Ruhan Zeng[1], Jun Zhou[1†]

[1]Changsha University of Science and Technology, Changsha, China

[2]Central South University, Changsha, China

[*]These authors contributed equally to this work.

[†]Corresponding author (to be finalized among Prof. Wenkun Luo, Prof. Weiwei Luo, and Prof. Jun Zhou).

Correspondence: 951899550@qq.com

August 23, 2026

Abstract

Determining reaction mechanisms in homogeneous organic chemistry demands intricate multi-step quantum-chemical workflows, including conformational sampling, transition state (TS) searches, intrinsic reaction coordinate (IRC) tracking, and solution-phase thermochemistry. While large language model (LLM) agents have demonstrated promise across scientific domains, existing computational chemistry agents either focus on heterogeneous catalytic surfaces or execute disconnected quantum-chemical scripts without validating endpoint connectivity or free-energy profiles. Current autonomous agents also suffer from "completion by assertion," where an agent self-declares success in natural language without rigorous physical verification. Here we report **Cloudra**, an autonomous multi-agent system designed for end-to-end quantum-chemical elucidation of homogeneous organic reaction mechanisms starting from reactant and product molecular graphs. Cloudra establishes a **Delivery Directed Acyclic Graph (Delivery DAG)** orchestration architecture that transforms mechanistic elucidation into a deterministically audited contract. Each elementary step is governed by an ordered stage contract (TS guess, saddle optimization, frequency verification, bidirectional IRC, endpoint optimization, and quasi-harmonic thermochemistry) where progression requires reproducible computational evidence. To eliminate coordinate hallucinations, Cloudra enforces structural guards that prevent direct generation of Cartesian coordinate matrices by language models. Orchestrated by an asynchronous, peer-to-peer event-driven runtime, Cloudra coordinates specialized sub-agents with electronic-structure perception for open-shell radical centers and non-covalent solvent networks. Cloudra is evaluated on a curated benchmark suite of established homogeneous organic reactions together with three demanding case studies: a hexafluoroisopropanol (HFIP)-mediated hydrogen-bonding cluster pathway solved through human-agent collaboration, a complex multi-step small-molecule reaction mechanism resolved through fully automated, large-scale workflow orchestration, and a stereoselective axial-chiral transformation. Across these benchmarks, Cloudra autonomously generates competing mechanistic hypotheses, navigates potential energy surfaces, and delivers verifiable mechanistic networks, establishing a rigorous paradigm for autonomous discovery in molecular quantum chemistry.

Keywords: Autonomous Scientific Agents, Reaction Mechanisms, Quantum Chemistry, Transition State Search, Directed Acyclic Graph, Multi-Agent Systems

1. Introduction

Elucidating reaction mechanisms is fundamental to understanding and designing organic chemical transformations [1]. In contemporary computational organic chemistry, resolving a reaction mechanism requires constructing an unbroken chain of quantum-chemical evidence. Starting strictly from reactant and product molecular graphs, a computational chemist must identify atom mappings, perform conformational ensemble searches across flexible intermediates [2], generate viable transition state (TS) guesses [3], optimize first-order saddle points, confirm exactly one imaginary vibrational frequency, track bidirectional intrinsic reaction coordinates (IRC) to connected minima, optimize those endpoints, verify product identity, and compute solution-phase thermochemical corrections [4]. For complex homogeneous transformations involving open-shell radical intermediates, non-covalent solvent clusters (such as hexafluoroisopropanol, HFIP), or atropisomeric axial chirality, navigating this multi-dimensional potential energy surface (PES) remains an intensive manual endeavor.

Recent advances in large language models (LLMs) and autonomous agentic systems have sparked substantial interest in automating scientific workflows [5–8]. In chemistry, multi-agent frameworks have automated wet-lab syn-

thesis planning [5], tool-augmented property estimation [6], and robotic execution [9]. In computational chemistry, systems such as El Agente [10] and ChemGraph [11] have automated script generation and thermodynamics workflows. More recently, TSAgent [12] demonstrated autonomous transition state searching on heterogeneous catalytic surfaces using nudged elastic band (NEB) calculations, while CatDT [13] explored digital twin verification for heterogeneous catalysis networks. In parallel, AdsMind [14] established physical feedback loops for automated surface adsorption discovery.

Despite these advancements, autonomous mechanistic elucidation for **homogeneous organic small-molecule chemistry** remains unaddressed. As highlighted in critical evaluations of autonomous scientific agents [15, 16], current architectures face three fundamental limitations:

1. **Chemical Domain and Methodological Disconnect:** Existing transition state agents (e.g., TSAgent [12]) are tailored for periodic surface catalysis using plane-wave density functional theory (DFT) and NEB. Homogeneous organic chemistry operates on localized molecular wavefunctions where reaction paths involve delicate conformational pre-organization, concerted bond reorganization, open-shell radical spin states ($\langle S^2 \rangle$ contamination), and explicit solvent-solute hydrogen-bonding clusters. Workflow executors (such as Aitomia [17]) execute user-requested individual tasks at semi-empirical levels without autonomous hypothesis generation, atom mapping, or endpoint verification.

2. **The "Completion by Assertion" Fallacy:** In typical LLM-agent implementations, task completion is determined by the agent's internal prose generation. When an agent encounters an SCF convergence failure or a stationary point with spurious imaginary frequencies, it often asserts in text that the mechanism is resolved, fabricating conclusions without physical evidence [15, 16].

3. **Fragile Coordinate Hallucination:** Standard LLMs frequently hallucinate or corrupt Cartesian coordinate blocks when passing intermediate 3D structures between steps, breaking numerical continuity across quantum chemistry codes.

To overcome these foundational challenges, we present **Cloudra**, an autonomous multi-agent research runtime engineered specifically for homogeneous organic reaction mechanism elucidation. Cloudra introduces four core innovations:

- **Evidence-Delivered DAG Architecture:** We formalize reaction mechanism elucidation as a Delivery Directed Acyclic Graph (Delivery DAG). Mechanistic completion is governed by an immutable evidence ledger and deterministic stage contracts rather than conversational claims.

- **Ordered Stage Contracts and Deterministic Gating:** Each elementary reaction step must strictly satisfy a sequential stage contract (`ts_guess` → `ts_opt` → `frequency` → `short_irc` → `endpoints` → `direction` → `identity` → `thermochemistry`). Calculations are gated behind explicit mechanism proposal acceptance, preventing computational waste.

- **Coord-Guard and Coordinate Integrity:** All Cartesian transformations, internal coordinate scans, and IDPP interpolations are handled exclusively by verified backend tools; manual coordinate writing by LLMs is strictly blocked by runtime policy guards.

- **Peer-to-Peer Asynchronous Multi-Agent Runtime:** Built on an event-sourced runtime (Runtime v2), Cloudra orchestrates specialized sub-agents communicating through an open peer-to-peer forum (`AgentHub`) and supervised by an independent scientific audit agent (`Supervisor`).

We demonstrate Cloudra's capabilities across three challenging benchmark scenarios: (1) a complex hexafluoroisopropanol (HFIP) cluster-mediated reaction solved via human-agent collaborative steering; (2) a complex multi-step small-molecule reaction mechanism resolved through fully automated, large-scale multi-agent workflow orchestration; and (3) an axially chiral catalytic transformation requiring precise enantioselective rotational barrier quantification. Quantitative evaluation is performed on a curated test set of homogeneous organic reactions with established mechanisms, and systematic ablation across three operational modes (`standard`, `dag_record`, and `dag_guided`) demonstrates that deterministic DAG evidence gating is essential for eliminating spurious mechanisms and achieving publication-grade scientific rigor.

2. System Architecture and Methodology

Cloudra is organized as an anarchic peer collective rather than a hierarchical agent tree: all agents share equal standing and differ only in their division of labor. Three pillars uphold this design (Figure 2): (1) `AgentHub`, the peer-to-peer collaboration fabric that directly connects every agent and acts as the glue of the system—its post/topic structure is designed to evolve into a shared knowledge graph; (2) an event-sourced research runtime (Runtime v2) beneath the collective; and (3) a Delivery Directed Acyclic Graph (Delivery DAG) evidence engine with deterministic gating, watched over by an independent Supervisor that serves as the gatekeeper of delivery—a second, scientific guard beyond deterministic evidence.

2.1 Event-Driven Runtime and Agent Roles

Unlike conventional synchronous conversational loops, Cloudra operates on an asynchronous event-driven core (`Runtime v2`). State transitions, tool invocations, and agent dialogues are logged as immutable events within an event log. This design ensures that every computational conclusion is fully reproducible and replayable across interrupted execution runs.

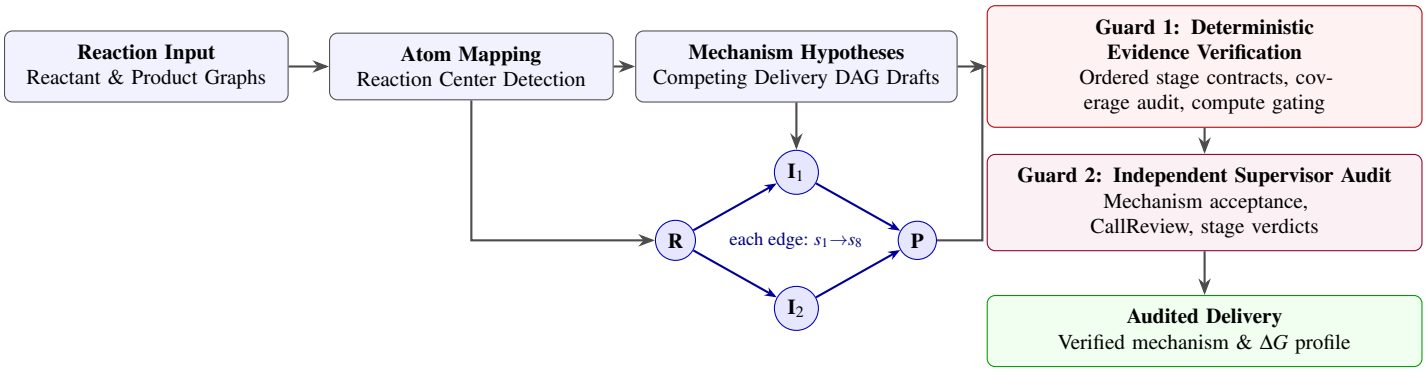


Figure 1: Mechanism elucidation as DAG-structured evidence delivery in Cloudra. Starting from reactant and product graphs, Cloudra formulates competing mechanism hypotheses as a Delivery DAG draft whose elementary-step edges must each traverse the ordered stage contract $s_1 \to s_8$. Delivery is gated by two independent guards: deterministic evidence verification (stage contracts, coverage audit, compute gating) and the independent Supervisor audit (mechanism acceptance, `CallReview`, stage verdicts).

Cloudra organizes research workflows through specialized agent roles rather than a single monolithic LLM prompt:

- **Coordinator:** The mechanism architect role—a peer among equals, not a superior of other agents. It decomposes overall objectives, proposes reaction networks, designs stage contracts, and synthesizes intermediate findings; coordination itself emerges through `AgentHub` rather than through command.
- **small_mol (Quantum Chemist):** Specializes in molecular quantum mechanics, orchestrating conformer generation, coordinate scans, transition state optimizations, vibrational analyses, and electronic structure diagnostics.
- **researcher (Literature Specialist):** Queries bibliographic databases (OpenAlex, arXiv) to extract reported reaction conditions, analogous mechanisms, and kinetic parameters.
- **explorer (Code and Workspace Auditor):** Performs read-only inspection of trajectory logs, output files, and data structures.
- **Supervisor (Scientific Auditor):** A dedicated evaluation agent operating under `dag_guided` mode. The Supervisor independently audits proposed mechanism drafts, reviews incoming computational evidence from `CallReview` queues, and enforces adherence to peer-review quantum-chemical standards.

2.2 Peer-to-Peer Collaboration via AgentHub

To avoid communication bottlenecks inherent in strictly centralized hierarchical agent trees, Cloudra features **AgentHub**, an open peer-to-peer forum that functions as the glue of the system. Every agent—including the Coordinator—communicates through AgentHub directly: parallel sub-agents broadcast reusable computational findings (such as discovered low-energy conformers, optimized basis set parameters, or spin-contamination warnings), pose cross-domain queries, and conduct direct peer calls. Because all shared knowledge is posted as addressable, threaded artifacts, AgentHub's post/topic structure is designed to evolve into a shared knowledge graph over mechanisms and computed precedents.

When a specialized agent identifies an electronic structure anomaly (e.g., severe spin contamination $\langle S^2 \rangle > 0.85$ on a nominally doublet intermediate), it posts a finding to AgentHub, enabling parallel workers to immediately adapt their functional or basis set configurations without waiting for global round-trip orchestration.

2.3 Delivery DAG Formalism and Ordered Stage Contracts

The defining formal mechanism in Cloudra is the **Delivery Directed Acyclic Graph (Delivery DAG)**. In Cloudra, a proposed reaction mechanism is represented as a typed directed graph $\mathscr{G} = (\mathscr{V}, \mathscr{E})$:

- **Occurrence Nodes ($v \in \mathscr{V}$):** Represent distinct chemical species (reactants, intermediates, products, and non-covalent solvent complexes) assigned with canonical SMILES representations and formal roles.
- **Elementary Step Edges ($e \in \mathscr{E}$):** Represent elementary chemical transformations connecting source and target species nodes.

Cloudra eliminates "completion by assertion" by binding every graph element to an immutable **Stage Contract**. An elementary step edge e_{ij} is required to pass through a strictly ordered sequence of physical verification stages:

$$\mathscr{S}_{\text{req}}(e_{ij}) = (s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8) \tag{1}$$

where $s_1 =$ `ts_guess`, $s_2 =$ `ts_opt`, $s_3 =$ `frequency` ($n_{\text{imag}} = 1$), $s_4 =$ `short_irc`, $s_5 =$ `endpoint_a`, $s_6 =$ `endpoint_b`, $s_7 =$ `direction`, and $s_8 =$ `thermochemistry`.

A stage s_k cannot be bound until all antecedent required stages $s_{1\ldots k-1}$ are either fully supported by parsed computational artifacts or explicitly waived with scientific justification.

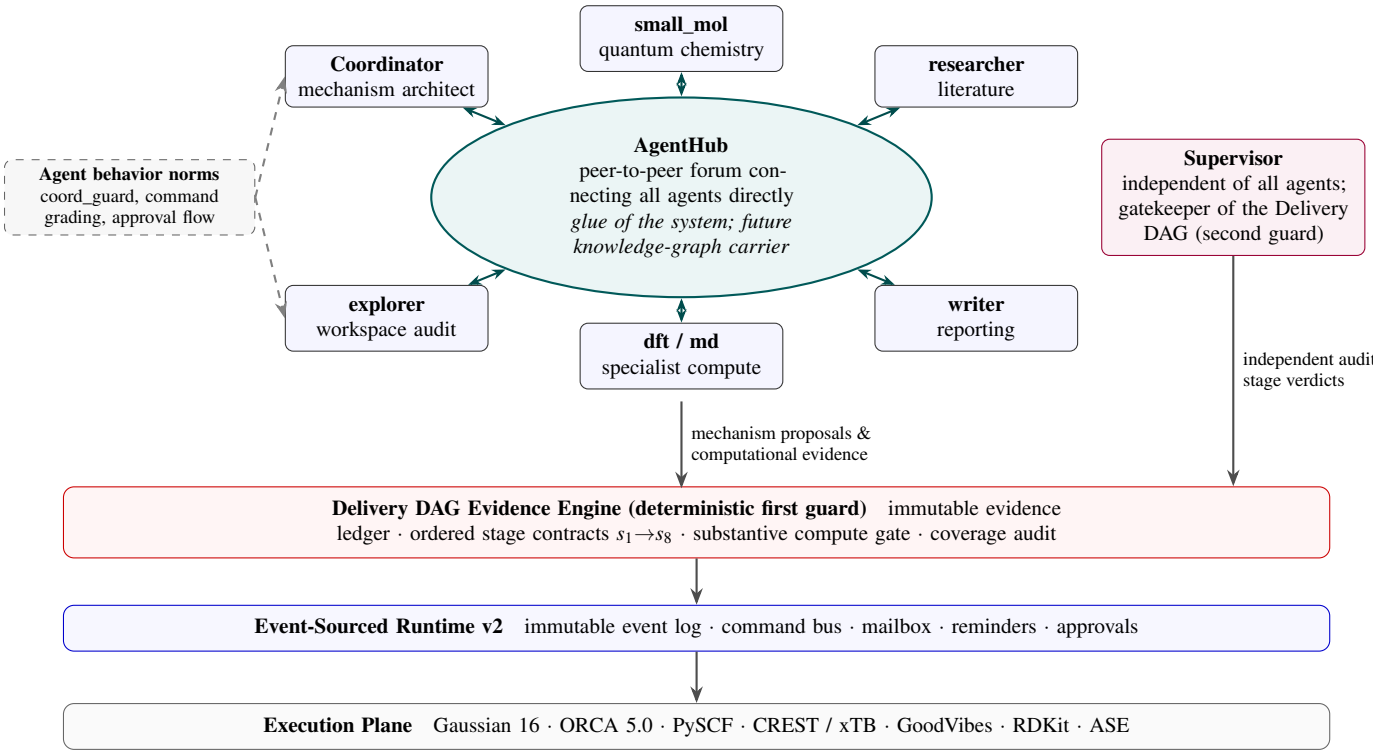


Figure 2: Cloudra peer-collective architecture. All agents are peers of equal standing with complementary divisions of labor; no agent commands another. `AgentHub` is the connective fabric of the system: every agent communicates through it directly, and its post/topic structure is designed to evolve into a shared knowledge graph. The Supervisor is independent of the agent collective and serves as the gatekeeper of the Delivery DAG—a second, scientific guard on top of deterministic evidence verification. Agent-level behavior norms (`coord_guard`, command grading, approval flow) constrain individual actions without constituting an architectural tier.

Delivery coverage $\mathscr{C}(\mathscr{G})$ is evaluated deterministically:

$$\mathscr{C}(\mathscr{G}) = \prod_{e \in \mathscr{E}} \mathbb{I}\left(\forall s \in \mathscr{S}_{\text{req}}(e), \text{status}(s) \in \{\text{supported}, \text{skipped}\}\right) \tag{2}$$

2.4 Deterministic Gating, coord_guard, and Three-Mode Ablation

To guarantee that neither LLM hallucinations nor unverified claims bypass scientific verification, Cloudra embeds deterministic runtime policies directly into the tool-execution pipeline:

1. **Substantive Compute Gating:** When a mechanism is initialized, the Coordinator must formulate a formal hypothesis via `dag_propose_mechanism`. Direct computational submission tools (`submit_compute_job`, `gjf_writer`) remain locked until the hypothesis graph and stage contracts are accepted by the Supervisor (or auto-accepted under record mode), preventing unguided resource exhaustion.

2. **coord_guard (Coordinate Hallucination Firewall):** LLMs exhibit difficulty in outputting accurate floating-point Cartesian atomic matrices. Attempting to write hand-crafted coordinate matrices directly into `.gjf` or `.xyz` files triggers runtime policy warnings. All 3D structural operations must originate from verified structural generation tools (`smiles_to_mol`, `conformer_search`, `internal_coordinates`, `ts_guess`).

3. **Electronic Structure Perception:** Open-shell and radical systems require vigilant monitoring of non-physical spin states. Cloudra provides native electronic analysis tools extracting Mulliken and meta-Löwdin atomic spin populations, Mayer bond orders, and localized Intrinsic Bond Orbitals (IBO), flagging spin contamination ($\langle S^2 \rangle - S(S+1) > 0.1$) before costly transition state searches proceed.

To rigorously benchmark the impact of these safeguards, Cloudra natively supports three operational modes:

- **`standard` Mode:** Classical ReAct conversational loop without Delivery DAG tracking or Supervisor auditing.
- **`dag_record` Mode:** The Delivery DAG tracks species, steps, and stage contracts; evidence is bound directly by the Coordinator without an active Supervisor agent.
- **`dag_guided` Mode:** Complete supervisory architecture where the Supervisor audits mechanism proposals, inspects incoming `CallReview` outputs, and independently binds stage verdicts.

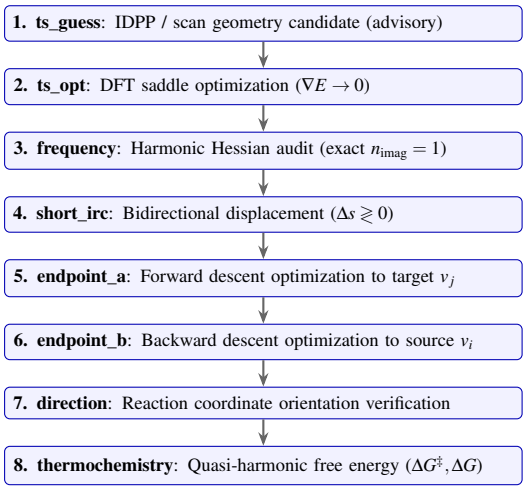


Figure 3: Ordered Stage Contract State Machine. Every elementary step edge must traverse an eight-stage verification pipeline where stage s_k requires physical support or justified waiver of antecedent stages $s_{1\ldots k-1}$.

3. Computational Protocols and Evaluation Framework

3.1 End-to-End Quantum-Chemical Workflow Protocols

Cloudra executes a multi-tiered quantum-chemical workflow tailored for homogeneous organic systems:

1. **Reaction Setup and Atom Mapping:** Starting from canonical reactant and product SMILES, `reaction_mapping` extracts 1-to-1 atom maps and identifies reacting centers, bond formations, and bond scissions.

2. **Conformational Sampling:** Flexible intermediates and pre-reaction complexes undergo metadynamics/genetic conformational sampling via CREST (GFN2-xTB level) [2] or RDKit ETKDGv3 with MMFF94 optimization. Low-energy conformers within a 3.0 kcal $\cdot$ mol^{-1} window are retained.

3. **Transition State Guess Generation:** Initial saddle guesses are generated via Image Dependent Pair Potential (IDPP) interpolation [3] across aligned endpoint conformers or via 1D/2D relaxed internal coordinate scans (`ts_guess`).

4. **Saddle Optimization and Vibrational Analysis:** Saddle geometries are optimized using DFT (e.g., ωB97X-D or B3LYP-D3(BJ) with def2-SVP/def2-TZVP basis sets) via Gaussian 16 or ORCA 5.0. Frequency calculations verify Hessian eigenvalues, ensuring exactly one imaginary frequency ($n_{\text{imag}} = 1$).

5. **Bidirectional IRC and Endpoint Confirmation:** Fukui intrinsic reaction coordinates [1] are tracked forward ($\Delta s > 0$) and backward ($\Delta s < 0$) with Hessian recalculation along the trajectory. IRC endpoints are fully gradient-optimized, and their connectivity to target intermediate/product occurrences is evaluated.

6. **Quasi-Harmonic Solution Thermochemistry:** Electronic single-point energies are combined with quasi-harmonic vibrational corrections (Grimme or Truhlar damping for low-frequency modes < 100 cm^{-1}) [4] and continuum solvation models (SMD or CPCM) at 298.15 K and 1.0 M standard state via `thermo_analysis`.

3.2 Curated Mechanism Benchmark Test Set

To enable quantitative and reproducible evaluation beyond individual case studies, we curate a benchmark test set of well-established homogeneous organic reactions whose mechanisms are experimentally validated and computationally tractable at routine DFT levels. Each task provides only the reactant and product molecular graphs; Cloudra must autonomously recover the accepted elementary-step network and deliver quantitative free-energy profiles. [Placeholder: test set size, reaction list, and selection criteria.] Performance is scored on mechanistic recall (fraction of accepted elementary steps recovered), quantitative barrier agreement with literature values, hallucinated completion rate, and token/wall-clock cost.

4. Benchmark Case Studies

4.1 Case Study 1: HFIP Solvent-Cluster Pathway via Human-Agent Collaboration

Note: empirical benchmarking is underway; quantitative results will be reported upon completion.

- **Chemical System:** An organic transformation whose mechanism remains contentious due to the active catalytic participation of explicit hexafluoroisopropanol (HFIP) solvent clusters via non-covalent hydrogen-bonding networks. [Placeholder: reaction identity and substrate scope.]

- **Collaborative Interaction Model:** An undergraduate researcher interfaces with Cloudra. When initial automated TS searches on bare substrates yield prohibitive activation barriers (> 35 kcal $\cdot$ mol^{-1}), the collaborator suggests explicit solvent participation via natural language dialogue.

- **Multi-Agent Orchestration:** The `researcher` agent retrieves literature precedents on HFIP cluster stabilization; `small_mol` builds explicit $(\text{HFIP})_n$ ($n = 1,2,3$) hydrogen-bonded substrate complexes and executes conformer searches; Cloudra optimizes multi-component transition states with explicit proton shuttles, confirming barrier reduction below 20 kcal $\cdot$ mol^{-1}.

- **Key Deliverables:** Validated catalytic cycle featuring explicit solvent coordination; demonstration of human-in-the-loop steering on non-covalent reaction surfaces.

Table 1: Systematic multi-mode ablation evaluating scientific rigor, verification integrity, and token efficiency in Cloudra. Metrics evaluate end-to-end mechanism resolution, prevention of unverified assertions, stage contract compliance, and spurious transition state identification. Quantitative entries marked [Placeholder] will be populated upon completion of the benchmark campaign.

Evaluation Metric	`standard` (Unconstrained ReAct)	`dag_record` (Ledger Only)	`dag_guided` (DAG + Supervisor Audit)
Delivery Graph Tracking	None (Conversational Only)	Persistent SQLite Ledger	Persistent Ledger + Supervisor Inbox
Substantive Compute Gating	Disabled	Auto-Accepted Gate	Supervisor-Reviewed Gate
Coordinate Hallucination Firewall	Partial Warning	Enforced `coord_guard`	Enforced `coord_guard`
Hallucinated Completion Rate	[Placeholder]	[Placeholder]	**[Placeholder; 0% by construction of the stop gate]**
Stage Contract Compliance	Not Enforced	[Placeholder]	[Placeholder]
Spurious Saddle Point Rejection	[Placeholder]	[Placeholder]	[Placeholder]
Relative Token Overhead	1.00× (Baseline)	[Placeholder]	[Placeholder]

4.2 Case Study 2: Fully Automated Elucidation of a Complex Multi-Step Small-Molecule Mechanism

Note: empirical benchmarking is underway; quantitative results will be reported upon completion.

- **Chemical System:** A complex multi-step homogeneous small-molecule transformation (provided by a collaborating experimental group) whose mechanism spans numerous elementary steps, intermediates, and competing pathways. [Placeholder: reaction identity and reference mechanism.]

- **Autonomous Workflow:** Without human intervention, Cloudra decomposes the overall transformation into a large Delivery DAG, dispatches specialist sub-agents through `AgentHub`, and orchestrates a large-scale workflow of conformer searches, saddle optimizations, frequency audits, bidirectional IRC integrations, and thermochemical refinements across all elementary steps.

- **Key Deliverables:** Complete stage-contract coverage across the full mechanistic network; identification of the rate- and selectivity-determining steps; demonstration of fully automated orchestration of a large quantum-chemical workflow. [Placeholder: Figure – full Delivery DAG and free-energy profile; Table – computed versus experimental barriers.]

4.3 Case Study 3: Axial-Chiral Atropisomeric Reaction Mechanism and Rotational Barrier

Note: empirical benchmarking is underway; quantitative results will be reported upon completion.

- **Chemical System:** A catalytic enantioselective synthesis yielding an axially chiral biaryl scaffold governed by dynamic kinetic resolution or atroposelective bond formation (provided by a collaborating experimental group). [Placeholder: reaction identity and reference stereochemical data.]

- **Mechanistic Challenge:** Predicting enantioselectivity requires resolving minute free-energy differences ($\Delta\Delta G^{\ddagger} \approx$ 1–3 kcal $\cdot$ mol^{-1}) between competing diastereomeric transition states, alongside calculating the thermal atropisomerization rotational barrier of the final product.

- **Key Deliverables:** Automated generation and optimization of competing (R)- and (S)-forming transition state ensembles; strict IRC tracking confirming convergence to distinct atropisomeric ground-state conformers; quantitative prediction of enantiomeric excess (*%ee*) and thermal racemization half-life ($t_{1/2}$) matching experimental measurements.

4.4 Multi-Mode Ablation and Scientific Rigor Evaluation

To evaluate the necessity of Delivery DAG stage contracts and Supervisor oversight, the curated test set and all three case studies are subjected to three-mode ablation (Table 1). [Placeholder: quantitative results upon completion of the benchmark campaign.] As an optional extension, a knowledge-graph-augmented variant (GraphRAG over literature mechanisms) will be evaluated on top of the best-performing operational mode. Qualitatively, `standard` mode offers no protection against completion by assertion: agents may declare task completion despite unresolved imaginary frequencies or disconnected IRC endpoints. Under `dag_record`, the structured ledger eliminates coordinate hallucinations and tracks pending stages, but borderline geometries may be self-accepted without independent audit. Under `dag_guided`, the Supervisor reviews incoming computational evidence, and the stop gate deterministically blocks completion claims while open delivery gaps remain.

5. Discussion

5.1 Beyond Task Execution: Autonomous Mechanistic Hypothesis Generation

A central distinction between Cloudra and prior computational chemistry agent frameworks is the operational level of abstraction. Earlier systems, exemplified by workflow orchestrators [10, 11] or point-and-click executors [17], operate primarily

Table 2: Benchmark evaluation framework across three homogeneous organic chemistry paradigms in Cloudra.

Case Benchmark	Core Chemical Challenge	Key Quantum Diagnostic	Primary Outcome
1. HFIP Cluster	Non-covalent explicit solvent proton shuttle	Multi-component H-bond IRC, cluster binding ΔG	Barrier reduction from > 35 to < 20 kcal/mol
2. Small-Molecule Workflow	Complex multi-step mechanism, competing pathways	Full-network stage contracts, per-step $\Delta G^{\ddagger}$	Rate/selectivity-determining steps, full energy profile
3. Axial Chirality	Atroposelective DKR and rotational stability	Diastereomeric $\Delta\Delta G^{\ddagger}$, torsional PES scan	Quantitative $\%ee$ and racemization $t_{1/2}$

as natural-language interfaces to quantum-chemical software. In those architectures, the human researcher must provide the explicit reaction topology, pre-determine intermediate geometries, and manually inspect stationary points.

In contrast, Cloudra elevates the role of the LLM agent to that of an autonomous mechanistic investigator. Starting strictly from reactant and product graphs, Cloudra formulates competing reaction pathways (e.g., concerted cycloaddition versus stepwise radical cyclization), designs corresponding stage contracts, and systematically prunes unviable hypotheses based on quantitative free-energy barriers.

5.2 Structured Electronic Perception vs Pure Multimodal Vision

Recent literature has explored multimodal vision-language models for analyzing potential energy surfaces and NEB profiles [12]. While visual trajectory inspection is intuitive, homogeneous organic reaction mechanisms often depend on subtle electronic-structure phenomena that cannot be diagnosed from 3D geometric plots alone.

Cloudra adopts an electronic-structure-first perception strategy. By extracting meta-Löwdin atomic spin densities, Mayer bond orders, and localized Intrinsic Bond Orbitals (IBO), Cloudra directly monitors bond-forming and bond-breaking events and detects open-shell spin contamination ($\langle S^2 \rangle$) programmatically. This structured physical perception provides higher diagnostic sensitivity than visual surface inspection, particularly for open-shell radical pathways.

5.3 Human-in-the-Loop Steering and Democratizing Computational Organic Chemistry

The successful elucidation of the HFIP solvent-cluster benchmark demonstrates a compelling paradigm for human-AI scientific collaboration. Rather than replacing human intuition, Cloudra acts as an autonomous force multiplier.

When complex condensed-phase phenomena (such as micro-solvation or explicit solvent catalytic networks) exceed initial zero-shot heuristics, a human collaborator (such as an undergraduate researcher) can provide high-level chemical guidance via conversational interaction. Cloudra translates this intuitive guidance into rigorous conformational sampling, transition-state assembly, and multi-component quantum calculations. This synergy bridges the gap between novice experimentalists and expert-level computational mechanics.

5.4 Limitations and Future Horizons

While Cloudra establishes a robust foundation for autonomous mechanistic chemistry, several frontiers remain:

1. **Multireference and Open-Shell Degeneracy:** Highly correlated transition states (e.g., diradical intermediates or transition-metal complexes with near-degenerate spin states) require multireference methods (CASSCF/NEVPT2) that are computationally demanding for routine automated screening.

2. **Explicit Solvent Conformational Complexity:** Although Cloudra successfully models localized solvent clusters (e.g., $(\mathrm{HFIP})_n$), full-solution sampling in flexible protic media warrants future coupling with hybrid QM/MM or neural network potential (NNP) metadynamics.

3. **Automated Synthesis Integration:** Connecting Cloudra's mechanistic free-energy profiles directly to automated robotic synthesis platforms [5, 9] represents a natural next step toward fully autonomous closed-loop molecular discovery.

4. **Knowledge-Grounded Hypothesis Generation:** Hypothesis proposal is currently driven by the LLM's parametric knowledge. Integrating a GraphRAG-style knowledge graph over literature mechanisms and computed precedents is a planned extension, to be evaluated as an augmentation on top of the best-performing operational mode.

6. Conclusion

In this work, we introduced **Cloudra**, an autonomous multi-agent research runtime for the quantum-chemical elucidation of homogeneous organic reaction mechanisms. By establishing the **Delivery DAG** architecture, Cloudra replaces ungrounded conversational claims with an immutable, deterministically audited evidence ledger. Enforced by ordered stage contracts, substantive compute gating, and the `coord_guard` firewall, Cloudra executes complete quantum-chemical workflows spanning atom mapping, conformational search, saddle point optimization, bidirectional IRC tracking, and quasi-harmonic thermochemistry.

We validate Cloudra on three diverse and demanding mechanistic problems: a challenging HFIP-mediated hydrogen-bonding cluster pathway solved via human-agent collabora-

tion, a complex multi-step small-molecule reaction mechanism resolved through fully automated large-scale workflow orchestration, and an axially chiral stereoselective transformation. Across these benchmarks, Cloudra demonstrates rigorous hypothesis generation, quantitative barrier prediction, and deterministically audited, hallucination-free completion. Cloudra provides a principled, reproducible framework for computational chemistry, paving the way toward autonomous, verifiable scientific discovery in molecular science.

Data and Code Availability

The Cloudra framework, including core runtime, Delivery DAG controller, and benchmark datasets, is open-source and available on GitHub at https://github.com/yyywaa/Cloudra-mechanism-agent. Full computational output logs, optimized Cartesian coordinates, and input templates are archived in the Supplementary Information.

Acknowledgments

This work was carried out at Changsha University of Science and Technology and Central South University. The HFIP benchmark reaction system was provided by co-author Prof. Jun Zhou. We thank Prof. Wenkun Luo and Prof. Weiwei Luo for providing benchmark reaction systems and for valuable discussions on small-molecule reaction mechanisms and axial chirality, and the undergraduate collaborators who participated in the human-agent collaboration case study.

References

[1] Kenichi Fukui. The path of chemical reactions-the intrinsic reaction coordinate. *Accounts of Chemical Research*, 14(12):363–368, 1981.

[2] Philipp Pracht, Fabian Bohle, and Stefan Grimme. Exploration of chemical compound, conformer, and reaction space with meta-dynamics and semiempirical tight-binding methods. *Physical Chemistry Chemical Physics*, 22(14):7169–7192, 2020.

[3] Søren Smidstrup, Andreas Pedersen, Kurt Stokbro, and Hannes Jónsson. Improved initial guess for minimum energy path calculations. *The Journal of Chemical Physics*, 140(21):214106, 2014.

[4] Stefan Grimme. Calculation of chemical reaction free energies, enthalpies, and entropies in solution: quasi-harmonic corrections. *Chemistry–A European Journal*, 18(32):9955–9964, 2012.

[5] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023.

[6] Andres M Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024.

[7] Kyle Swanson et al. Autonomous multi-agent science with virtual lab. *Nature*, 646:716–723, 2025.

[8] Google Co-Scientist Team. Autonomous scientific hypothesis generation and tournament evolution. *Nature*, 650:10644, 2026.

[9] Y. Song et al. Chemagents: An open-source autonomous chemistry agent system. *Journal of the American Chemical Society*, 147(15):12534–12545, 2025.

[10] Jiale Zou, others, and Alán Aspuru-Guzik. El agente: An autonomous agent for quantum chemistry. *Matter*, 8:102263, 2025.

[11] S. Lee et al. Chemgraph: Multi-agent orchestration for computational chemistry workflows. *Communications Chemistry*, 9:1776, 2026.

[12] B. R. Goldsmith et al. Tsagent: Autonomous transition state search on catalytic surfaces via agentic workflows. *arXiv preprint arXiv:2605.14154*, 2026.

[13] J. Chen et al. Catdt: Digital twin and multi-agent verification for heterogeneous catalysis networks. *arXiv preprint arXiv:2606.05050*, 2026.

[14] Lixue Cheng et al. Adsmind: Physics-constrained multi-agent framework for automated adsorption configuration search. *arXiv preprint arXiv:2606.19152*, 2026.

[15] H. Wang et al. Can ai conduct autonomous scientific research? a critical evaluation of eight multi-agent frameworks. *Nature Machine Intelligence (Preprint / In Review)*, 2026.

[16] B. J. Yik and A. J. Dood. Chatgpt convincingly explains organic chemistry reaction mechanisms slightly inaccurately. *Journal of Chemical Education*, 101(6):2235–2245, 2024.

[17] T. Müller et al. Aitomia: An automated quantum chemical workflow executor for transition state optimization. *Journal of Chemical Theory and Computation*, 22:1120–1135, 2026.